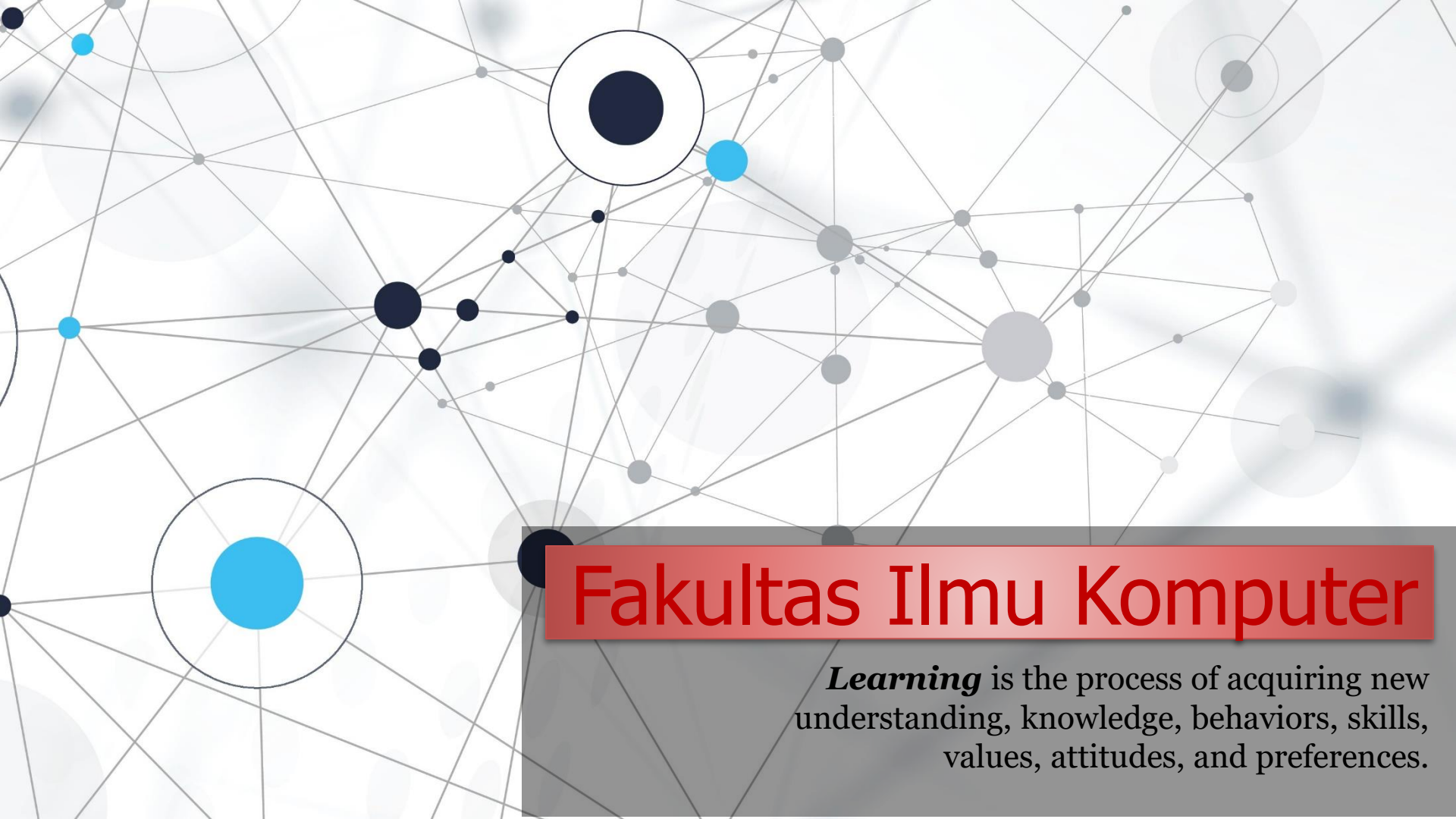




Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Security TLS & SSL (HTTP to HTTPS)
(Transport Layer Security & Secure Sockets Layer)



Welcome to *the* **DevOps**
Engineering *course*

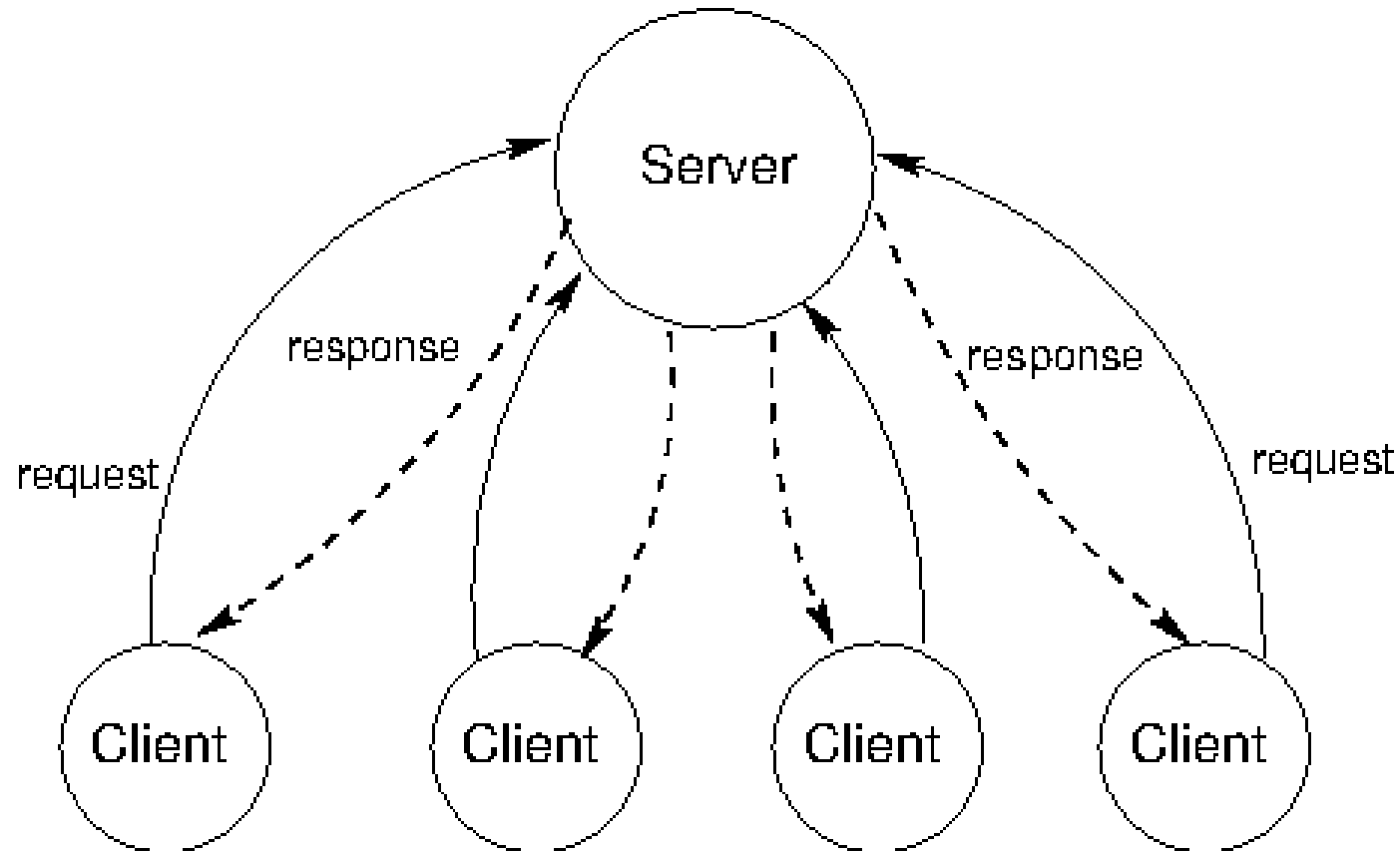
A decorative border of green leaves and branches at the top of the slide.

1 - TLS/SSL Implementation (HTTP *to* HTTPS)

A decorative border of green leaves and branches at the bottom of the slide.

Web Client & Web Server Communication

A **web client** is an application that communicates with a web server, using Hypertext Transfer Protocol (HTTP). The basic objective of the web server is to store, process and deliver web pages to the users. **Web browser is an example of a web client.** Basically, a **web server is a software that runs continuously and responds to requests** by other pieces of software, usually called a client.

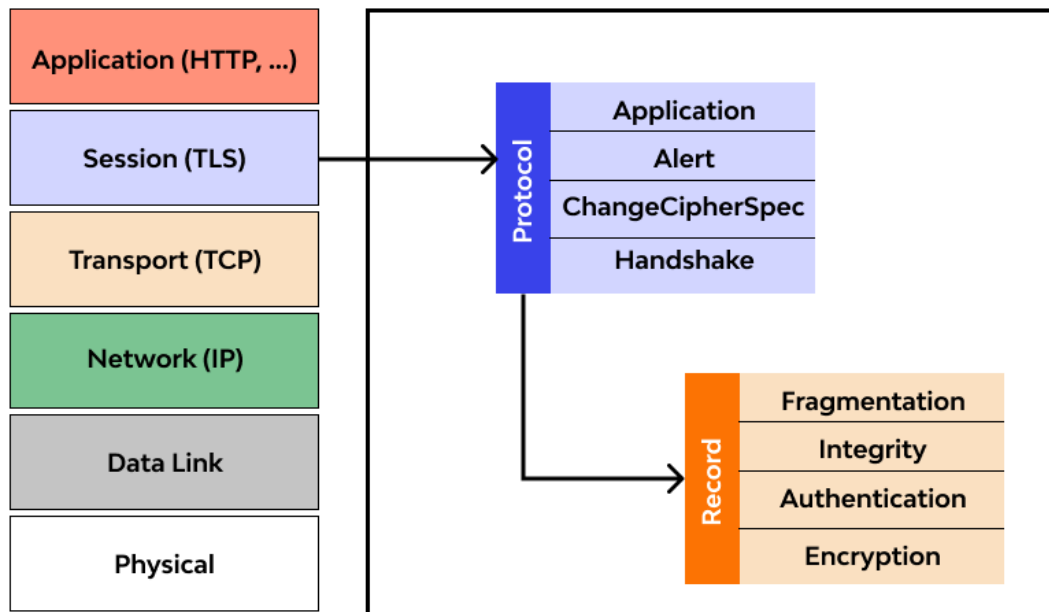




Security in the context of HTTP and HTTPS protocols, will inevitably involve the use of TLS (Transport Layer Security) and SSL (Secure Sockets Layer). The **implementation of TLS/SSL (HTTP to HTTPS) is essential to improve the security of web communication (client and server)**, especially when sensitive or private data must be transferred between the browser and the server, such as *login information, credit card details, customer data, government data* and others.

TLS (Transport Layer Security)

- ❖ Transport Layer Security (TLS) is a **cryptographic protocol designed to provide communications security** over a computer network. The protocol is widely used in applications such as email, instant messaging, and voice over IP, but its use in securing HTTPS remains the most publicly visible.
- ❖ TLS in web communication is a security protocol that **provides encryption and authenticity of data transferred over a network**. TLS replaces SSL and is designed to provide an additional layer of security on top of transport protocols such as TCP (Transmission Control Protocol).
- ❖ **TLS in web communication ensures that the data transferred between the user device and the server remains private and cannot be modified by unauthorized parties**. It involves the process of exchanging encryption keys and establishing a security session.



SSL (Secure Sockets Layer)

- ❖ SSL (Secure Sockets Layer) is an encryption-based Internet security protocol. It was first developed by Netscape in 1995 for the purpose of ensuring privacy, authentication, and data integrity in Internet communications.
- ❖ SSL is the predecessor (**pendahulu TLS**) to the modern TLS encryption used today. SSL in web communications was **the predecessor of TLS** and was used to provide similar security in transferring data over a network.
- ❖ SSL in web communications uses symmetric and asymmetric cryptography to protect data and provide trust in the identity of the server and, in some cases, in the client.



Secure Sockets Layer

HTTP to HTTPS

HTTP (Hypertext Transfer Protocol) is a protocol used to transfer data on the web. However, HTTP does not provide adequate security (*tidak menyediakan keamanan yang memadai*) because the **data transmitted between the web browser and the server is not encrypted**, making it vulnerable to attacks and monitoring by third parties.

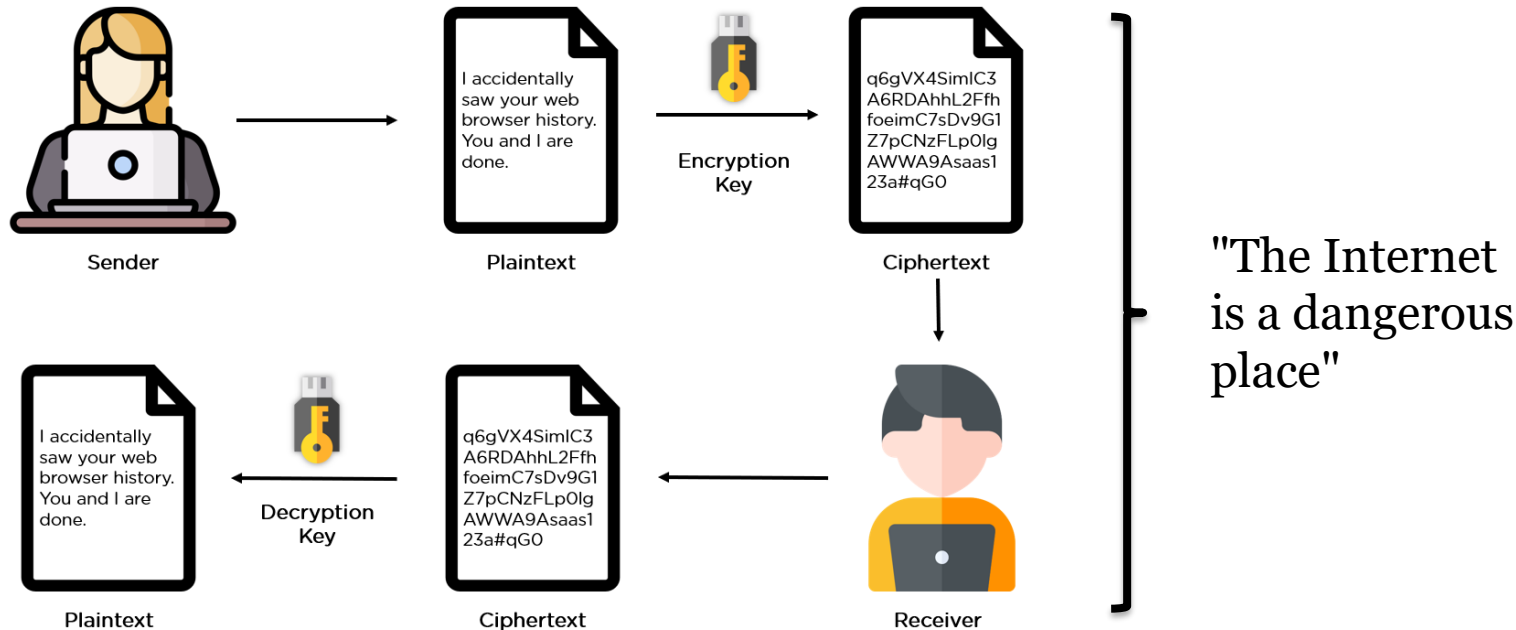


HTTPS (Hypertext Transfer Protocol Secure) is a secure version of HTTP. In HTTPS, the **data transferred between the web browser and the server is encrypted** using a security protocol, such as TLS or SSL, to protect the confidentiality and **integrity of the data**.



Mengapa komunikasi ini perlu dijaga?

Many websites today, especially those that provide **financial services**, **e-commerce**, or **collect personal information**, invest substantial resources into maintaining the security. These systems will undoubtedly use security protocols such as HTTPS to encrypt communication between users and servers, implement firewalls and intrusion detection systems, and regularly conduct security audits.



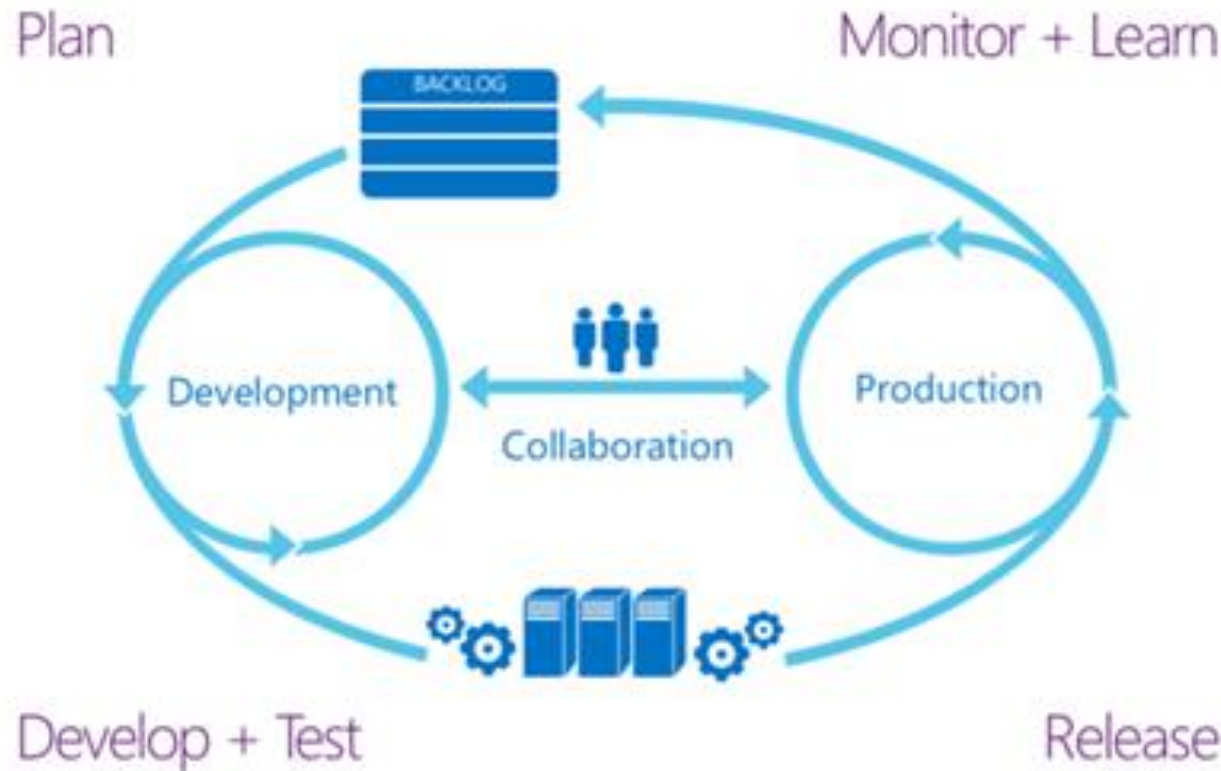
Internet dapat menjadi lingkungan yang berisiko karena adanya potensi ancaman keamanan seperti malware, phishing, and other cyber crimes. Selain itu, internet juga dapat menjadi tempat di mana informasi pribadi dapat terancam keamanannya (compromised).

A decorative border of green leaves and branches at the top of the slide.

1 – Why “HTTP *to* HTTPS” important in DevOps?

A decorative border of green leaves and branches at the bottom of the slide.

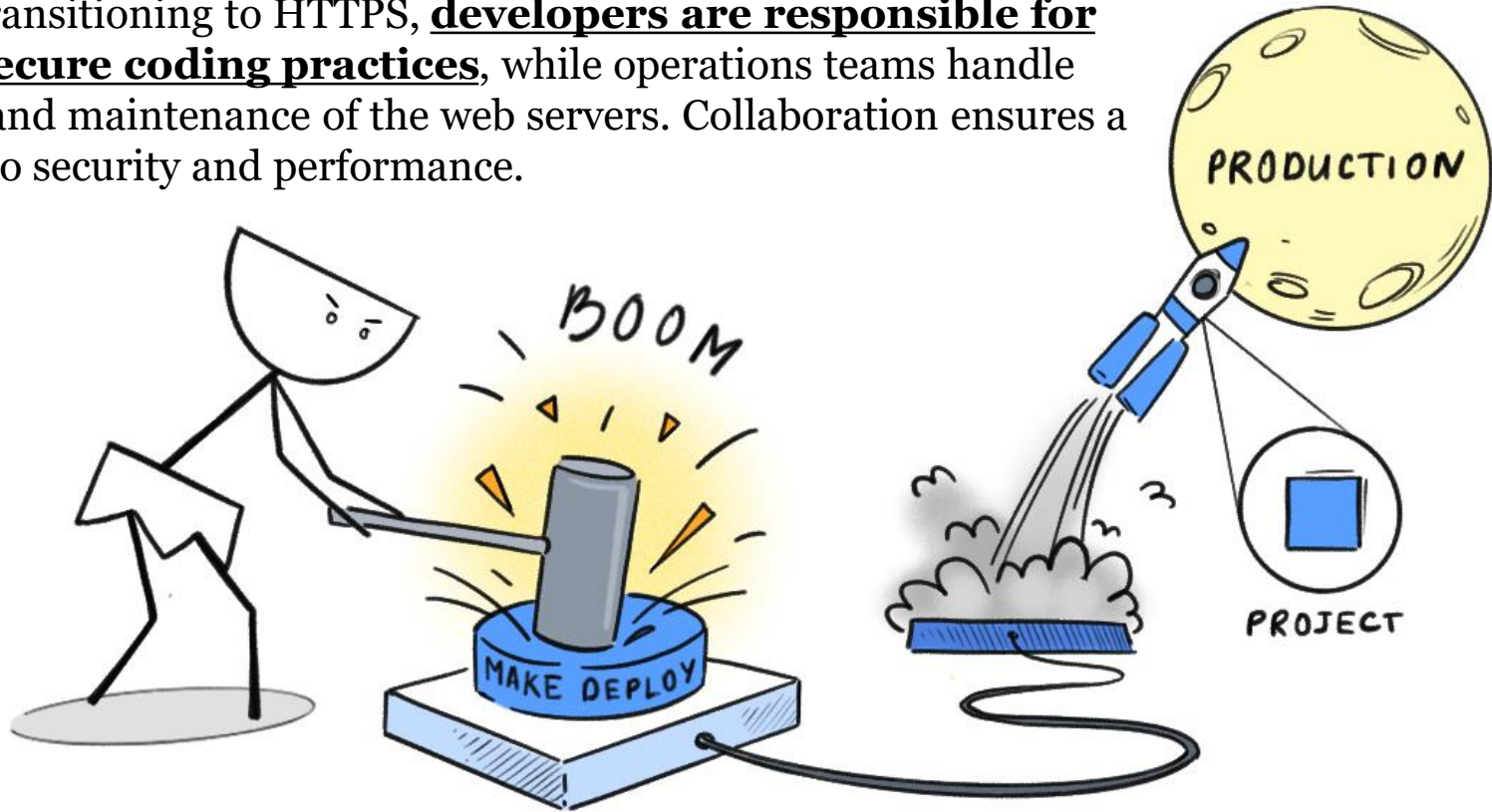
HTTP *to* HTTPS Involves *in* DevOps



The relationship between DevOps and the transition from HTTP to HTTPS involves ***collaboration***, ***automation***, and a focus on security and performance. DevOps practices play a crucial role in ensuring a smooth and secure transition, from planning and development to deployment and ongoing operations.

Collaboration between Dev & Ops

DevOps encourages collaboration between development and operations teams. In the context of transitioning to HTTPS, **developers are responsible for implementing secure coding practices**, while operations teams handle the configuration and maintenance of the web servers. Collaboration ensures a holistic approach to security and performance.



DevOps practices often **emphasize security as an integral part of the development lifecycle**. Transitioning from HTTP to HTTPS is a security measure to encrypt data in transit, providing a secure communication channel between the web client and server. DevOps teams may be involved in implementing and managing the HTTPS configuration to ensure compliance with security standards.



website



https://www.

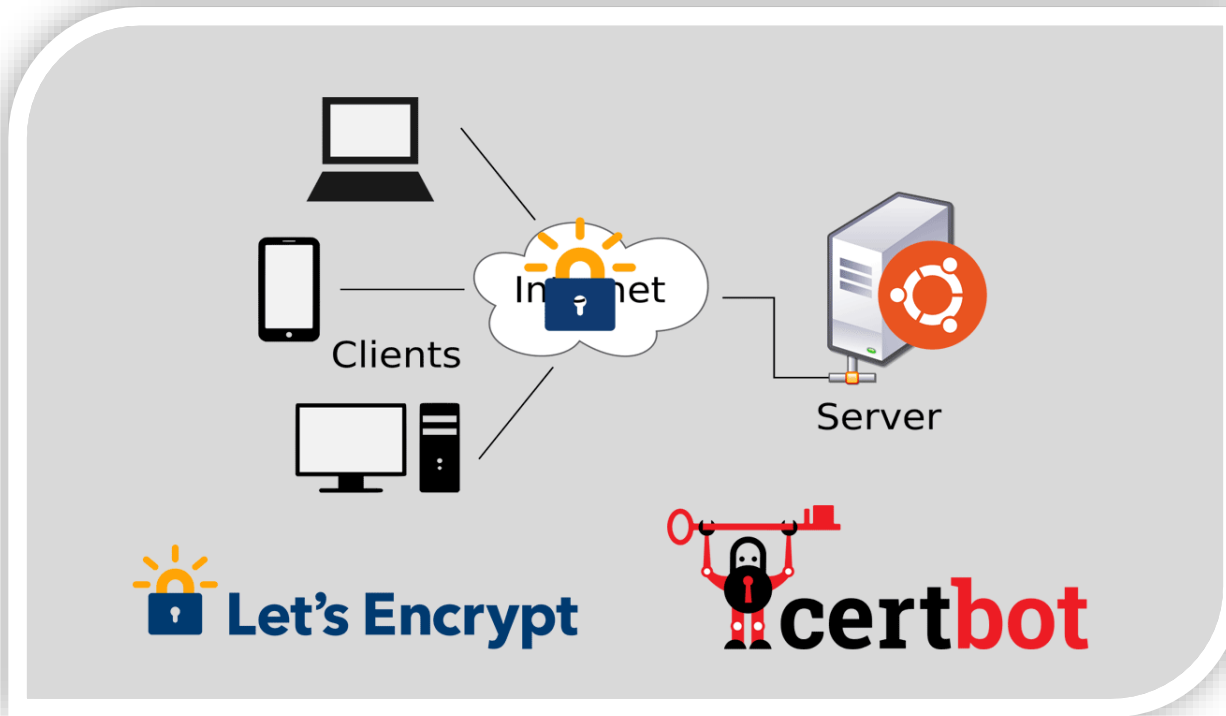


Connection is secure

■ Certificate (Valid)

SSL certificates (Secure Sockets Layer) are what enable websites to use HTTPS, which is more secure than HTTP. An SSL certificate is a data file hosted in a website's origin server. SSL certificates make SSL/TLS encryption possible, and they contain the website's public key and the website's identity, along with related information.

Let's Encrypt SSL Certificate



Let's Encrypt is a free, automated, and open certificate authority. Let's Encrypt is a global Certificate Authority (CA) to let people and organizations around the world obtain, renew, and manage SSL/TLS certificates. These certificates can be used by websites to enable secure HTTPS connections. Let's Encrypt offers Domain Validation (DV) certificates.

Common Requirement SSL Certificates

- 1) Requires a **DOMAIN NAME** to secure with an SSL certificate. Let's Encrypt provides certificates for public domain names that are accessible over the internet.
- 2) Requires access to the **SERVER** where the domain is hosted. This can be SSH access or any other type of access that allows us to manage the server configuration.
- 3) Need to have a **WEB SERVER** running on the server. Apache, Nginx, and several other web servers are supported by Let's Encrypt.
- 4) Need to have **CERTBOT (ACME CLIENT)** or another ACME (Automatic Certificate Management Environment) client installed on the server. Certbot is the official tool supported by Let's Encrypt to automate SSL certificate management. Ensure that the server's operating system supports Certbot and the latest version of Let's Encrypt. Ubuntu 22.04 should support Certbot well.



Exercise *for* Students



Exercise #1

(Buat domain name murah di niagahoster.co.id)

Open & Sign In NiagaHoster

<https://www.niagahoster.co.id/domain-murah>

The screenshot shows the NiagaHoster website interface. At the top, there's a navigation bar with the NiagaHoster logo, links for Hosting, VPS, Website Builder, and Domain, and a Log in button. A banner at the top right says 'PROMO BLACK FRIDAY TERBATAS: AMBIL PROMO >'. The main heading is 'Cek Domain Murah dan Beli Sekarang'. Below this, a subtext says 'Cek nama domain yang Anda inginkan, dan temukan domain impian Anda di sini'. There are two buttons: 'Cari Domain Baru' and 'Buat Domain dengan AI'. A search bar contains the text 'Cari nama domain yang Anda inginkan'. To the right of the search bar is a 'Cek Sekarang' button. Below the search bar, there are six boxes showing domain extensions and their prices: .id (Rp252.000 / Rp210.900), .com (Rp199.000 / Rp147.900), .online (Rp549.000 / Rp15.900), .shop (Rp520.000 / Rp14.900), .info (Rp819.000 / Rp62.900), and .xyz (Rp219.000 / Rp31.900). At the bottom, it says 'Tersedia proteksi privasi domain secara gratis' with a question mark icon.

Cek Domain Murah | Daftar Lang... x

niagahoster.co.id/domain-murah

PROMO BLACK FRIDAY TERBATAS: [AMBIL PROMO >](#)

NIAGAHOSTER

Hosting VPS Website Builder Domain Log in

Cek Domain Murah dan Beli Sekarang

Cek nama domain yang Anda inginkan, dan temukan domain impian Anda di sini

Cari Domain Baru ✨ Buat Domain dengan AI

Cari nama domain yang Anda inginkan

Cek Sekarang

.id Rp252.000 Rp210.900	.com Rp199.000 Rp147.900	.online Rp549.000 Rp15.900	.shop Rp520.000 Rp14.900	.info Rp819.000 Rp62.900	.xyz Rp219.000 Rp31.900
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Tersedia proteksi privasi domain secara gratis ?

Cari Domain Murah

Cek Domain Murah dan Beli Sekarang

Cek nama domain yang Anda inginkan, dan temukan domain impian Anda di sini!

[Cari Domain Baru](#) [Buat Domain dengan AI](#)

[Cek Sekarang](#)



Cari “semmywebsite.online”

Cek Domain Murah | Daftar Lan...

niagahoster.co.id/domain-murah

NIAGAHOSTER

Hosting VPS Website Builder Domain Log in

Hasil Pencarian Semua Ekstensi

Selamat! Domain yang Anda Cari Tersedia

semmywebsite.online **HEMAT 97%**

Rp549.900
Rp15.900/thn

[Pilih](#)

This domain is suitable for creating a personal or professional website, as it conveys a sense of being a semi-official or semi-formal online presence.

Click button ini untuk memilih domain name.

Durasi Registrasi Domain

Cart | Hostinger

cart.hostinger.com/pay/5368a8b7-8b98-4673-af8-d0f87d5f3d

Relaunch to update

NIAGAHOSTER Login

Selesaikan pembayaran

1. Pilih durasi paket

semmywebsite.online

Registrasi Domain .ONLINE
Registrasi diperpanjang seharga Rp549.900/thn pada 27/11/2024

Durasi

- 1 tahun
- 1 tahun
- 2 tahun
- 3 tahun

HEMAT 97%

Rp15.900
~~Rp549.900~~

GRATIS fitur perlindungan privasi domain

Rp0

2. Buat akun

Metode Pembayaran

Cart | Hostinger

cart.hostinger.com/pay/5368a8b7-8b98-4673-af8-d0f87dc55f3d

Relaunch to update

3. Pilih pembayaran

☒ Kartu Kredit

☐ BCA VA

☐ Mandiri Virtual Account

☐ BNI Virtual Account

☐ BRI Virtual Account

☐ CIMB Niaga Virtual Account

☐ Maybank Virtual Account

☐ Permata Bank Virtual Account

☐ QRIS

☐ GoPay

☐ OVO

☐ Indomaret

semmywebsite.online

✓ Registrasi domain - 1 tahun

Rp549.900 **Rp15.900**

✓ Proteksi Privasi Domain WHOIS

Rp0

Nama depan (opsional)

Nama belakang (opsional)

Nomor Telepon (opsional)

Negara

Indonesia

Kabupaten/provinsi (opsional)

Kota (opsional)

Alamat (opsional)

Kode pos (opsional)

Diskon Paket - 97%

-Rp534.000

Pajak & Biaya Tambahan

Rp4.868

Total

Rp554.768 **Rp20.768**

Punya Kode Kupon?

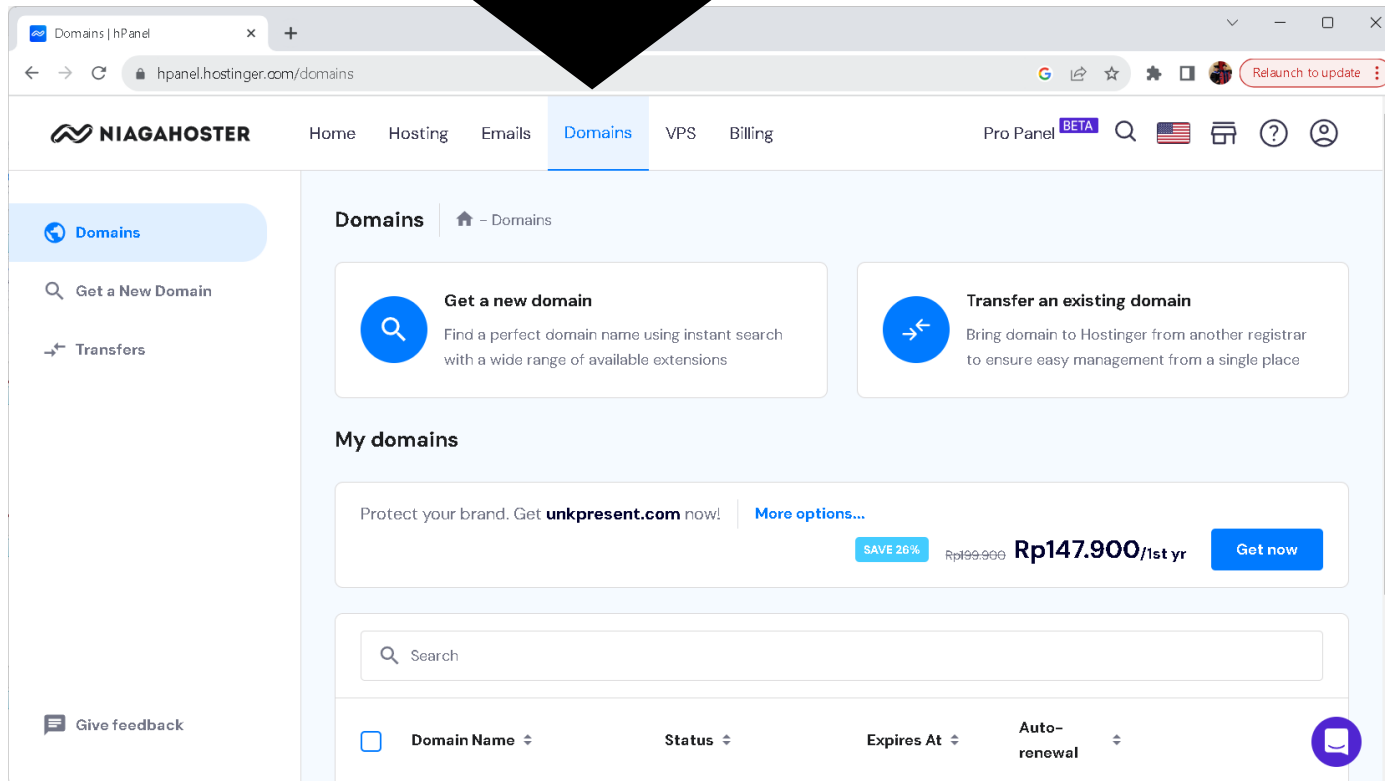
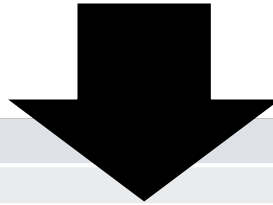
Buat Tagihan Pembayaran

Pembayaran Aman dan Terenkripsi

Dengan checkout, Anda setuju dengan [Ketentuan Penggunaan](#) kami dan mengonfirmasi bahwa Anda telah membaca [Kebijakan Privasi](#) kami. Anda dapat membatalkan biaya perpanjangan layanan kapan saja.

Masuk Ke Dashboard Niagahoster

Click di tab Domains



The screenshot shows the Niagahoster dashboard with the 'Domains' tab selected. The dashboard includes a sidebar with 'Domains' and 'Transfers' options. The main content area features two cards: 'Get a new domain' and 'Transfer an existing domain'. Below these is a section for 'My domains' with a search bar and a table header. The browser address bar shows 'hpanel.hostinger.com/domains'.

Domains | hPanel

hpanel.hostinger.com/domains

NIAGAHOSTER Home Hosting Emails **Domains** VPS Billing

Pro Panel **BETA** 🔍 🇺🇸 📁 ? 👤 Relaunch to update

Domains 🏠 - Domains

Get a new domain
Find a perfect domain name using instant search with a wide range of available extensions

Transfer an existing domain
Bring domain to Hostinger from another registrar to ensure easy management from a single place

My domains

Protect your brand. Get **unkpresent.com** now! [More options...](#)

SAVE 26% ~~Rp199.900~~ **Rp147.900/1st yr** [Get now](#)

🔍 Search

☐	Domain Name ▾	Status ▾	Expires At ▾	Auto-renewal ▾	📄
--------------------------	---------------	----------	--------------	----------------	---

Give feedback

Cek Status Domain Name

The screenshot shows the hPanel Domains management interface. A red arrow points to the 'Domains' menu item in the top navigation bar. Another red arrow points to the 'Manage' button in the bottom right corner of the domain list. A third red arrow points to the 'Active' status of the domain 'unkpresent.site'.

Domains | hPanel

hpanel.hostinger.com/domains

Home Hosting Emails **Domains** VPS Billing

Pro Panel BETA

Domains

Get a New Domain

Transfers

Get a new domain

Find a perfect domain name using instant search with a wide range of available extensions

Transfer an existing domain

Bring domain to Hostinger from another registrar to ensure easy management from a single place

My domains

Protect your brand. Get **unkpresent.com** now! [More options...](#) **SAVE 26%** ~~Rp199.999~~ **Rp147.900/1st yr** [Get now](#)

Search

☐	Domain Name	Status	Expires At	Auto-renewal	
☐	unkpresent.site	Active	2024-08-08	☐	Manage

Give feedback

Click button "Manage"

Pastikan Sudah Verified by Email

The screenshot shows the hPanel interface for a domain named 'unkpresent.site'. The 'Domains' tab is selected in the top navigation bar. The 'Domain Information' section displays the following details:

Domain Information	
Status	Active
Email verification status	Verified
Expires at	2024-08-08 Renew
Transfer authorization code	***** View
Nameservers	ns1.dns-parking.com ns2.dns-parking.com Change

Below the domain information, there are two sections:

- Auto-renewal**: A toggle switch is currently **DISABLED**. Below it, a text line reads "We will automatically renew the domain to keep it active" and a button labeled [Enable auto-renewal](#) is present.
- Forward your domain to an existing website**: A text line reads "All of your traffic will be redirected to your entered website." and a button labeled [Forward domain](#) is present.

In the bottom left corner, there is a [Give feedback](#) link. In the bottom right corner, there is a chat icon.

Click Tab “DNS/Nameservers”

The screenshot displays the Niagahoster hPanel interface for managing DNS records for the domain `unkpresent.site`. The left sidebar contains navigation links: **Domain Overview**, **DNS / Nameservers** (the active tab), and **Domain Ownership**. The main content area is titled **Manage DNS records** and includes a descriptive paragraph: "These records define how your domain behaves. Common uses include pointing your domain at web servers or configuring email delivery for your domain." Below this is a form to add new records with fields for **Type** (set to A), **Name** (set to @), **Points to**, and **TTL** (set to 14400), followed by an **Add Record** button. A table below the form lists existing DNS records with columns: **Type**, **Name**, **Priority**, **Content**, **TTL**, **Delete**, and **Edit**. The table contains two records, both of type A, with names `www` and `@`, priority 0, and content `27.112.78.66`. A large red arrow points to the **Content** column of the table, indicating where to change the IP address.

unkpresent.site

Domain Overview

DNS / Nameservers

Domain Ownership

Manage DNS records

These records define how your domain behaves. Common uses include pointing your domain at web servers or configuring email delivery for your domain.

Type: A, Name: @, Points to: , TTL: 14400, Add Record

Type	Name	Priority	Content	TTL		
A	www	0	27.112.78.66	14400	Delete	Edit
A	@	0	27.112.78.66	14400	Delete	Edit

Type, Name, Priority, Content, TTL

Change with your public IP Address



ASSIGNMENT

Exercise #2

(Install Let's Encrypt SSL Certificate on Ubuntu 22.04)

Install Certbot Client

1) Updates the package lists:

➤ **sudo apt-get update**

2) Install the **Certbot client** with the Apache plugin:

➤ **sudo snap install --classic certbot**

➤ `sudo apt-get install certbot python3-certbot-apache`

➤ `sudo apt-get install certbot --fix-missing`

➤ `sudo apt-get install certbot python3-certbot-apache apache2`

3) Test if the Certbot client was installed correctly:

➤ **sudo certbot --help**

4) Obtain certificate for domain name:

➤ **sudo certbot --apache**

Sesudah perintah ini dijalankan, anda akan diminta untuk memasukkan email dan domain name yang telah dibuat.

Renew Certificate (after 3 months)

1) Certbot is on Autorenewal:

- `sudo systemctl list-timers`
- `sudo systemctl enable certbot.timer`
- `sudo systemctl start certbot.timer`
- `sudo systemctl status certbot.timer`
- `sudo certbot renew --dry-run`

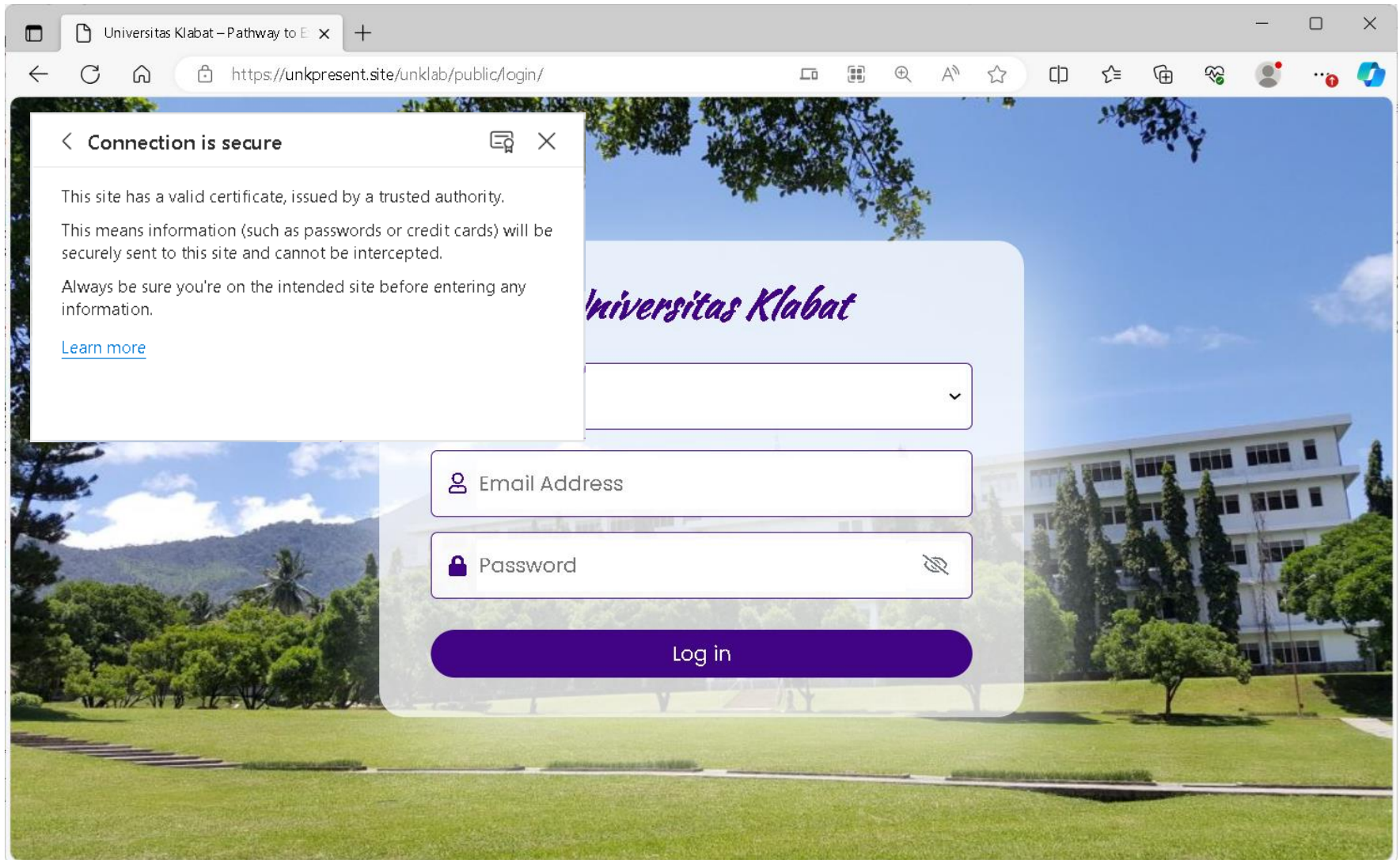
2) Cek Status Certificate:

- `sudo certbot certificates`

3) Renew Certificate:

- `sudo certbot renew`

Check Https Access



END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

